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Centre Number

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Student Number

SCEGGS Darlinghurst

2005

**Higher School Certificate
Trial Examination**

Mathematics Extension 1

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1–7
- All questions are of equal value

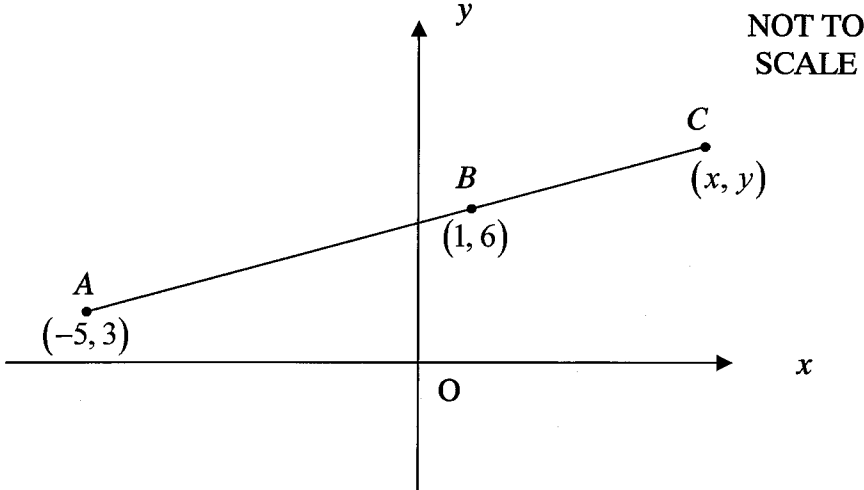
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Total marks – 84

Attempt Questions 1–7

All questions are of equal value

Answer each question in a SEPARATE writing booklet

- | | Marks |
|---|--------------|
| Question 1 (12 marks) | |
| (a) Find $\frac{d}{dx}(\tan^{-1} 2x)$ | 2 |
| (b) Find the obtuse angle between the two straight lines $y = x - 1$ and $2x + y = 1$. Answer correct to the nearest degree. | 2 |
| <hr/> | |
| (c) Evaluate $\int_0^{\frac{\pi}{6}} \sec 2x \tan 2x \, dx$ | 3 |
| (d) | 2 |
| <div style="text-align: center;"></div> | |
| Given that $AC : CB = 5 : 2$, find the co-ordinates of the point C. | |
| (e) Use the substitution $u = x^2 - 6x + 7$ to find the exact value of | 3 |

$$\int_0^1 \frac{x-3}{x^2-6x+7} \, dx .$$

Question 2 (12 marks) Begin a NEW writing booklet

- (a) How many different positive integers can be formed from the digits 1, 3, 5, 7 if a digit cannot be used more than once in a particular number? 2

- (b) Consider $P(x) = 2 + 3x - 3x^2 - 2x^3$

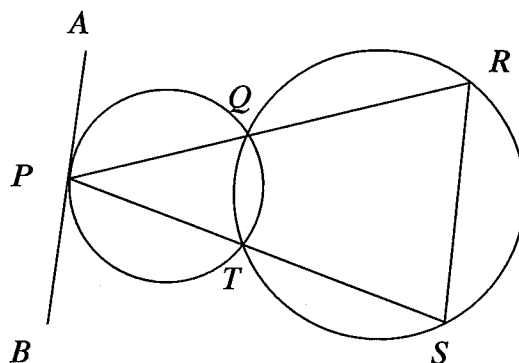
- (i) Prove $P(1) = 0$. 1

- (ii) Solve $P(x) \leq 0$. 3

- (c) Find the term independent of x in the expansion of 3

$$\left(2x^2 - \frac{1}{2x}\right)^6$$

- (d) 3

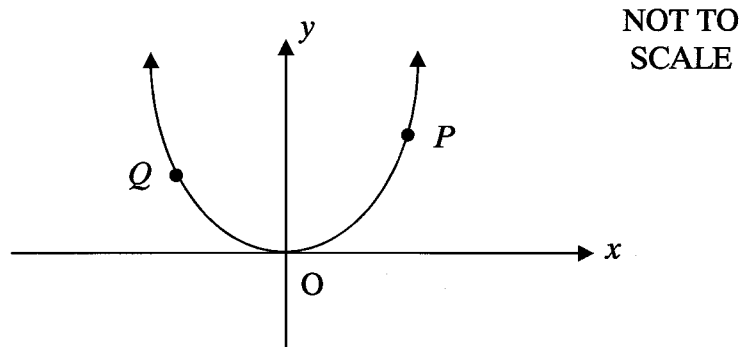


The two circles intersect at Q and T .
 AB is a tangent to the smaller circle at P .
 PQR and PTS are straight lines.

Prove that the tangent at P is parallel to the chord RS .

Question 3 (12 marks) Begin a NEW writing booklet

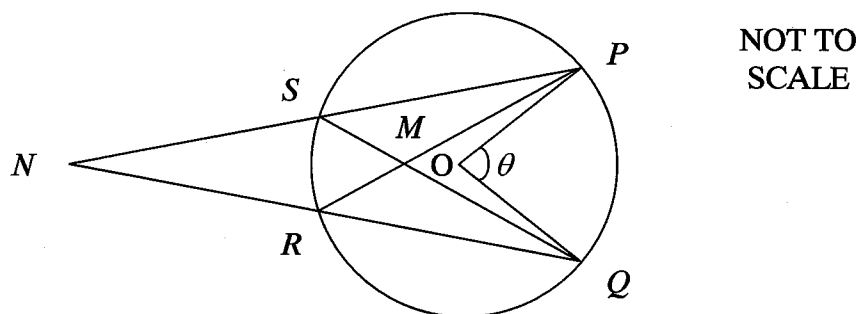
- (a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$ with vertex $(0, 0)$ as shown below.



- | | | |
|-------|---|---|
| | (i) Find the equation of the tangent to the parabola at P . | 1 |
| <hr/> | | |
| | (ii) Hence, prove that the tangents at P and Q intersect at the point $R(a(p+q), apq)$. | 3 |
| | (iii) State the condition that the tangents intersect on the directrix. | 1 |
| (b) | (i) Prove that there is a solution to the equation $2\sin \frac{\pi}{2}x - 2x + 3 = 0$ between $x = 1.5$ and $x = 2$ where x is measured in radians. | 2 |
| | (ii) Using an initial approximation of $x = 1.75$ and one application of Newton's Method, find a better approximation correct to 4 significant figures. | 2 |
| (c) | Find the exact volume formed when the region bounded by $y = 1 + \sin \frac{x}{2}$, the x axis and $x = 0$ to $x = \frac{\pi}{2}$ is rotated about the x axis. | 3 |

Question 4 (12 marks) Begin a NEW writing booklet

(a)



In the diagram, P, Q, R and S are points on the circle centre O . $\angle POQ = \theta$
The straight lines PS and QR intersect at N and PR and QS intersect at M .

(i) Prove $\angle PRN = 180^\circ - \frac{1}{2}\theta$ 1

(ii) Prove $\angle PMQ + \angle PNQ = \theta$ 2

(b) (i) Express $\sin 2x - 2\cos 2x$ in the form $A\sin(2x - \alpha)$ for $A > 0$ and $0 \leq \alpha \leq 90^\circ$. 2

(ii) Hence solve $\sin 2x - 2\cos 2x = 1$ for $0 \leq x \leq 180^\circ$. 2

(c) Consider $f(x) = \frac{2x}{x-1}$:

(i) Sketch the hyperbola $y = f(x)$ showing important features. 2

(ii) Find $y = f^{-1}(x)$. 1

(iii) State the domain and range of $y = f^{-1}(x)$. 2

Question 5 (12 marks) Begin a NEW writing booklet

(a) Prove that $\int_0^1 \frac{dx}{\sqrt{4-3x^2}} = \frac{\pi}{3\sqrt{3}}.$ 3

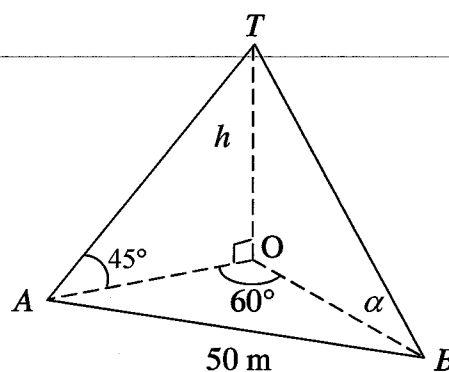
(b) Use Mathematical Induction to prove that 4

$$\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{n}{2^n} = 2 - \frac{n+2}{2^n}$$

for an integer $n > 0$.

(c)

NOT TO
SCALE



In the diagram, the points A, B and O are in the same horizontal plane.
A and B are 50m apart and $\angle AOB = 60^\circ$. OT is a vertical tower of height h metres.
The angles of elevation of T from A and B respectively are 45° and α . (α is acute.)

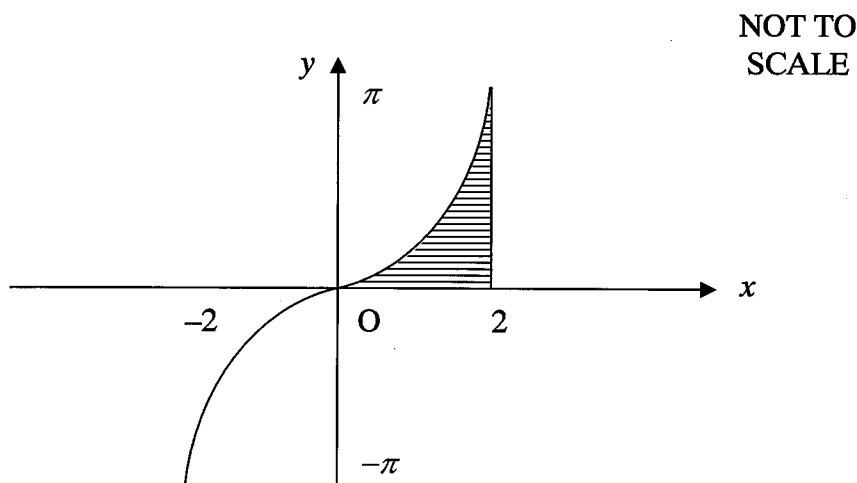
(i) Prove $AO = h$. 1

(ii) Prove $h^2 \cot^2 \alpha - h^2 \cot \alpha + h^2 = 50^2$ 2

(iii) Given the tower is 30 m high, find the angle α correct to the nearest degree. 2

Question 6 (12 marks) Begin a NEW writing booklet

- (a) The curve shown is $y = 2\sin^{-1} Bx$.



- | | |
|---|---|
| (i) Evaluate B . | 1 |
| (ii) Find the exact value of the shaded area. | 3 |
- (b) A ball is thrown at ground level with an initial velocity $V\text{ms}^{-1}$ and an angle of projection of α with the horizontal.
- You may assume the equations of motion.
- | | |
|---------------------------|----------------------------------|
| $\ddot{x} = 0$ | $\ddot{y} = -10$ |
| $\dot{x} = V \cos \alpha$ | $\dot{y} = -10t + V \sin \alpha$ |
| $x = Vt \cos \alpha$ | $y = -5t^2 + Vt \sin \alpha$ |
- | | |
|--|---|
| (i) Prove that the horizontal range is $\frac{V^2}{10} \sin 2\alpha$. | 2 |
| (ii) Explain why the maximum horizontal range occurs when $\alpha = 45^\circ$. | 1 |
| (iii) Find the maximum horizontal range where $V = 30\text{ms}^{-1}$. | 1 |
| (iv) How much further can the ball be thrown under these conditions if it is projected from a platform 10m above the ground? | 4 |

Question 7 (12 marks) Begin a NEW writing booklet

- (a) Use the substitution
- $u = \tan x$
- to evaluate

4

$$\int_0^{\frac{\pi}{4}} \frac{dx}{9\cos^2 x + 25\sin^2 x}$$

- (b) Given
- $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$
- ,

prove:

$$(i) \quad \binom{n}{1} + 3\binom{n}{2} + 9\binom{n}{3} + \dots + 3^{n-1}\binom{n}{n} = \frac{1}{3}(2^{2n} - 1) \quad 2$$

where n is a positive integer.

$$(ii) \quad \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} = 2^{n-1} \quad 2$$

where n is an even integer.

- (c) A class of 20 students consists of 12 girls and 8 boys. For a discussion session, 4 students are chosen at random to form a committee. The committee then chooses 1 of these 4 students at random to be the chairman.

How many of these committees:

- (i) have 4 female members? 1
- (ii) have at least 1 male member? 1
- (iii) have a male chairman? 2

End of Paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Extension 1 Trial 2005.

① a) $\frac{d}{dx}(\tan^{-1} 2x) = \frac{2}{1+4x^2}$

b) $m_1 = 1, m_2 = -2$

if θ acute, $\tan \theta = \left| \frac{1+2}{1-2} \right| = 3$.

$\theta = 72^\circ$ (nearest degree)

$\therefore$ obtuse angle is 108° (nearest degree)

c) $\int_0^{\frac{\pi}{6}} \sec 2x \tan 2x \cdot dx = \frac{1}{2} [\sec 2x]_0^{\frac{\pi}{6}}$

$= \frac{1}{2} (\sec \frac{\pi}{3} - \sec 0)$

$= \frac{1}{2} (2-1) = \frac{1}{2}$

d) External division 5:-2.

$x = \frac{-5x-2+1 \times 5}{3} \quad y = \frac{3x-2+6 \times 5}{3}$

$= 5 \quad = 8$

C is (5, 8)

e) $\frac{du}{dx} = 2x-6$, if $x=1, u=2$
 $x=0, u=7$.

$\therefore \int_0^1 \frac{x-3}{x^2-6x+7} \cdot dx = \int_7^2 \frac{x-3}{u} \cdot \frac{du}{2x-6}$

$= \frac{1}{2} \int_7^2 \frac{du}{u}$

$= \frac{1}{2} [\log_e u]_7^2$

$= \frac{1}{2} \log_e \frac{2}{7}$

② a) one digit numbers = 4

two " " = $4 \times 3 = 12$

three " " = $4 \times 3 \times 2 = 24$

four " " = $4 \times 3 \times 2 \times 1 = 24$

$\therefore$ Total number = 64.

Look at formula sheet

Careful of the signs in the formula.

Look at formula sheet

Well done.

Watch where new limits go.

$\int \frac{1}{u} du = \log_e u + c$

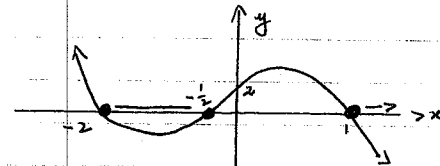
Numbers have a max^m of 4 digits
So can have 1, 2, 3 or 4 digits.

b) (i) $P(1) = 2+3-3-2 = 0$

(ii)

$$\begin{array}{r} -2x^2 - 5x - 2 \\ x-1 \overline{) -2x^3 - 3x^2 + 3x + 2} \\ \underline{-2x^3 + 2x^2} \\ -5x^2 + 3x \\ \underline{-5x^2 + 5x} \\ -2x + 2 \\ \underline{-2x + 2} \\ 0 \end{array}$$

$\therefore P(x) = (x-1)(-2x^2-5x-2)$
 $= -(x-1)(2x^2+5x+2)$
 $= -(x-1)(2x+1)(x+2)$



Solution: $-2 \leq x \leq -\frac{1}{2}$ and $x \geq 1$

c) $T_{k+1} = \binom{6}{k} (2x^2)^{6-k} \left(-\frac{1}{2}x^{-1}\right)^k$
 $= \binom{6}{k} 2^{6-k} \left(-\frac{1}{2}\right)^k x^{12-2k-k}$

$\therefore$ if independent of x , $12-3k=0$
 $k=4$

Term is $\binom{6}{4} 2^2 \times \left(-\frac{1}{2}\right)^4 = \frac{15}{4}$

d) Construction: join AT

Proof: $\angle QPA = \angle PTA$ ($\angle$ between tangent and chord = $\angle$ in alternate segment)
 $\angle PTA = \angle QRS$ (ext. $\angle$ of cyclic quad. = interior opposite $\angle$)

$\therefore \angle APQ = \angle QRS$

But these angles are in the alternate position with PR transversal. $\therefore AB \parallel RS$.

Well done.

A 'natural' follow on from part (i)

A clue that this is the shape of the graph is that the y -intercept is 2.

Check your zeroes are correct.

Know the formula

Well done.

Careful: $\angle APR \neq \angle RSP$.

$$a) (i) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

at P, gradient is $\frac{4ap}{4a} = p$.

equation of tangent is: $y - ap^2 = p(x - 2ap)$
 $= px - 2ap^2$

$\therefore px - y - ap^2 = 0$ is tangent at P.

(i) tangent at Q: $qx - y - aq^2 = 0$

Subtracting: $px - qx = ap^2 - aq^2$

$$(p - q)x = a(p - q)(p + q)$$

$$x = a(p + q)$$

$$y = pa(p + q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$= apq$$

$\therefore R$ is $(a(p + q), apq)$

(i) Directrix is $y = -a$.

$$\therefore apq = -a$$

condition is $pq = -1$

(i) Let $P(x) = 2\sin \frac{\pi}{2}x - 2x + 3$

$$P(1.5) = 2\sin \frac{\pi}{2} \times 1.5 - 3 + 3 = 1.4142 \dots$$

$$P(2) = 2\sin \frac{\pi}{2} \times 2 - 4 + 3 = -1$$

Since the sign has changed and the function is continuous there is a solution between $x = 1.5$ and 2

$$(ii) P(1.75) = 2\sin \frac{\pi}{2} \times 1.75 - 3.5 + 3 = 0.265366 \dots$$

$$P'(x) = 2 \times \frac{\pi}{2} \cos \frac{\pi}{2}x - 2$$

$$P'(1.75) = \pi \cos \frac{\pi}{2} \times 1.75 - 2 = -4.90245 \dots$$

must find the gradient at P!

no need to do this again, just put q for p .

This is the condition. There are other properties

Radians!

must say continuous

differentiate carefully.

$$\text{new estimate} = 1.75 - \frac{0.2653 \dots}{-4.902 \dots}$$

$$= 1.804 \text{ (4 s.f.)}$$

$$c) V = \pi \int_0^{\frac{\pi}{2}} (1 + \sin \frac{x}{2})^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} 1 + 2\sin \frac{x}{2} + \sin^2 \frac{x}{2} dx$$

$$= \pi \int_0^{\frac{\pi}{2}} 1 + 2\sin \frac{x}{2} + \frac{1}{2} - \frac{1}{2} \cos x dx$$

$$= \pi \left[\frac{3x}{2} - 4\cos \frac{x}{2} - \frac{1}{2} \sin x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\frac{3\pi}{4} - 4\cos \frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} - 0 + 4\cos 0 + \frac{1}{2} \sin 0 \right]$$

$$= \pi \left[\frac{3\pi}{4} - \frac{4}{\sqrt{2}} - \frac{1}{2} + 4 \right]$$

$$\text{Volume} = \pi \left[\frac{3\pi}{4} - 2\sqrt{2} + \frac{7}{2} \right] \text{ m}^3$$

$$(4) (i) \angle PRQ = \frac{1}{2} \angle POR \text{ (angle at centre is twice the angle at the circumference subtended by the same arc)}$$

$$= \frac{1}{2} \theta$$

$$\angle PRN = 180^\circ - \frac{1}{2} \theta \text{ (straight angle } \angle PRQ \text{ is } 180^\circ)$$

$$(ii) \text{ Similarly } \angle NSM = 180^\circ - \frac{1}{2} \theta$$

$$\angle PNQ + \angle SMR + \angle PRN + \angle NSM = 360^\circ \text{ (angle sum of quad. is } 360^\circ)$$

$$\therefore \angle PNQ + \angle SMR + 360^\circ - \theta = 360^\circ$$

$$\therefore \angle PNQ + \angle SMR = \theta$$

$$\text{but } \angle SMR = \angle PMR \text{ (vertically opp. angles are =)}$$

$$\therefore \angle PNQ + \angle SMR = \theta$$

$$b) (i) A \sin(2x - \alpha) = A \sin 2x \cos \alpha - A \cos 2x \sin \alpha$$

$$= \sin 2x - 2 \cos 2x$$

$$\therefore A \cos \alpha = 1 \quad \text{and} \quad A \sin \alpha = 2$$

Watch y^2 for volume.

This question required care and attention to small details.

well done.

Don't give up.

Try to find more information so marks can be allocated.

$$A = \sqrt{4+1} = \sqrt{5} \quad (A > 0)$$

$$\cos \alpha = \frac{1}{\sqrt{5}}$$

$$\alpha = 63^\circ 26' \text{ (nearest minute) (acute)}$$

$$\therefore \sin 2x - 2 \cos 2x = \sqrt{5} \sin(2x - 63^\circ 26')$$

$$(ii) \therefore \sin(2x - 63^\circ 26') = \frac{1}{\sqrt{5}}$$

$$\text{Acute angle} = 26^\circ 34', \quad 1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ quadrants}$$

$$\therefore 2x - 63^\circ 26' = 26^\circ 34', \text{ or } 153^\circ 26'$$

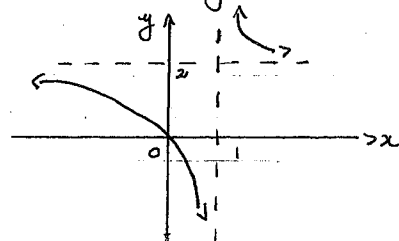
$$2x = 90^\circ, \quad 216^\circ 52'$$

$$x = 45^\circ, \quad 108^\circ 26'. \text{ (nearest minute)}$$

$$c) (i) f(x) = \frac{2x}{x-1} = \frac{2x-2+2}{x-1}$$

$$y = 2 + \frac{2}{x-1}$$

$$\text{asymptotes: } x=1, y=2.$$



$$(ii) y = \frac{2x}{x-1}$$

$$\therefore x = \frac{2y}{y-1}$$

$$xy - x = 2y$$

$$y(x-2) = 2$$

$$y = \frac{x}{x-2}$$

$$\therefore f^{-1}(x) = \frac{x}{x-2}$$

(iii) Domain:

all real x except $x=2$

Range:

all real y except $y=1$

well done.

well done.

Problems with horizontal asymptote.

line passes through (0,0)

(ii) must rewrite until you have $y = \dots$

(iii) the reverse of (i) which gives a clue to the graph of (i).

$$(5) a) \int_0^1 \frac{dx}{\sqrt{4-3x^2}} = \frac{1}{\sqrt{3}} \int_0^1 \frac{dx}{\sqrt{\frac{4}{3}-x^2}}$$

$$= \frac{1}{\sqrt{3}} \left[\sin^{-1} \frac{\sqrt{3}}{2} x \right]_0^1$$

$$= \frac{1}{\sqrt{3}} \left(\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \right)$$

$$= \frac{1}{\sqrt{3}} \times \left(\frac{\pi}{3} - 0 \right)$$

$$= \frac{\pi}{3\sqrt{3}}$$

well done.

b) Consider $n=1$,

$$\text{L.H.S.} = \frac{1}{2} \quad \text{R.H.S.} = 2 - \frac{3}{2} = \frac{1}{2}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{i.e. Assume } \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{k}{2^k} = 2 - \frac{k+2}{2^k}$$

Consider $n=k+1$

$$\text{L.H.S.} = \frac{1}{2} + \frac{2}{2^2} + \dots + \frac{k}{2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \frac{2(k+2) - k - 1}{2^{k+1}} \quad (\text{note signs here!})$$

$$= 2 - \frac{2k+4-k-1}{2^{k+1}}$$

$$= 2 - \frac{k+3}{2^{k+1}} = 2 - \frac{(k+1)+2}{2^{k+1}}$$

$$= \text{R.H.S. if } n=k+1$$

$\therefore$ If true for $n=k$, the statement is true for $n=k+1$. But it is true for $n=1$, and thus is true for $n=2, 3, \dots$ etc. i.e. true for all n integer > 0 .

Too many errors with negative signs.

Knowing the answer needed then trying to achieve it by dubious means is really obvious.

$$(i) \tan 45^\circ = \frac{h}{AO} = 1$$

$$\therefore AO = h$$

$$(ii) \tan \lambda = \frac{h}{OB}$$

$$\therefore OB = h \cot \lambda$$

using cosine rule:

$$AB^2 = AO^2 + OB^2 - 2 \times AO \times OB \times \cos \angle AOB$$

$$50^2 = h^2 + h^2 \cot^2 \lambda - 2 \times h \times h \cot \lambda \times \cos 60^\circ$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\therefore h^2 \cot^2 \lambda - h^2 \cot \lambda + h^2 = 50^2$$

$$(iii) 900 (\cot^2 \lambda - \cot \lambda + 1) = 2500$$

$$\therefore 900 \cot^2 \lambda - 900 \cot \lambda - 1600 = 0$$

$$9 \cot^2 \lambda - 9 \cot \lambda - 16 = 0$$

$$\cot \lambda = \frac{9 \pm \sqrt{81 + 576}}{18}$$

$$= \frac{9 \pm 25.632}{18}$$

$$\cot \lambda = \frac{34.632}{18} \quad (\lambda \text{ is acute})$$

$$\lambda = 27^\circ \text{ (nearest degree)}$$

$$2) a) (i) B = \frac{1}{2}$$

$$(ii) y = 2 \sin^{-1} \frac{x}{2}$$

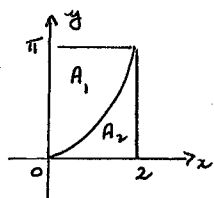
$$\frac{y}{2} = \sin^{-1} \frac{x}{2}$$

$$\therefore \frac{x}{2} = \sin \frac{y}{2}$$

$$x = 2 \sin \frac{y}{2}$$

$$A_1 = \int_0^\pi 2 \sin \frac{y}{2} dy$$

$$= 2 \left[-2 \cos \frac{y}{2} \right]_0^\pi$$



cannot integrate $\sin^{-1} x$ at Extension 1 level. So you must find area between curve and y axis. Then subtract from the area of rectangle

✓ well done.

well done.

Don't be afraid of the quadratic formula.

Explain answers that are discarded.

well done.

$$A_1 = -4 \left(\cos \frac{\pi}{2} - \cos 0 \right)$$

$$= 4u^2$$

$$\therefore \text{Shaded area} = (2\pi - 4)u^2$$

b) (i) Horizontal range is x when $y=0$ for the second time.

well done.

$$t(-5t + V \sin \lambda) = 0$$

$$t = 0 \text{ or } \frac{V \sin \lambda}{5} \quad (t=0 \text{ is initial value})$$

$$\text{horizontal range} = \frac{V \cos \lambda \times V \sin \lambda}{5}$$

$$= \frac{V^2 \times 2 \sin \lambda \cos \lambda}{10}$$

$$= \frac{V^2 \sin 2\lambda}{10}$$

(ii) maximum value of $\sin 2\lambda$ is 1

$\therefore$ maximum value for horizontal range is $\frac{V^2}{10}$ which occurs when $2\lambda = 90^\circ$ i.e. when $\lambda = 45^\circ$ (λ acute)

Explain carefully.

(iii) maximum range is $\frac{30^2}{10} = 90\text{m}$

well done.

(iv) If projected 10m above ground; $V=30$, $\lambda=45^\circ$

$$\ddot{x} = 0$$

$$\dot{x} = \frac{30}{\sqrt{2}}$$

$$x = \frac{30}{\sqrt{2}} t$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + \frac{30}{\sqrt{2}}$$

$$y = -5t^2 + \frac{30}{\sqrt{2}} t + 10$$

if $y=0$,

$$5t^2 - \frac{30}{\sqrt{2}} t - 10 = 0$$

$$t^2 - \frac{6}{\sqrt{2}} t - 2 = 0$$

$$t = \frac{\frac{6}{\sqrt{2}} \pm \sqrt{18+8}}{2}$$

Quadratic formula again

$$t = 4.6708301 \text{ or } -0.4281 \dots$$

$$\text{but } t \geq 0, \quad x = \frac{30}{\sqrt{2}} \times 4.6708301$$

$$= 99.08 \dots \text{ m.}$$

$\therefore$ can be thrown approximately 9m further.

7) a) $u = \tan x$ if $x = \frac{\pi}{4}$, $u = 1$
 $\frac{du}{dx} = \sec^2 x$ $x = 0$, $u = 0$.

$$\begin{aligned} dx &= \cos^2 x \, du. \\ \int_0^{\frac{\pi}{4}} \frac{dx}{9\cos^2 x + 25\sin^2 x} &= \int_0^1 \frac{\cos^2 x \, du}{9\cos^2 x + 25\sin^2 x} \quad \leftarrow \begin{array}{l} \div \cos^2 x \\ \div \cos^2 x \end{array} \\ &= \int_0^1 \frac{du}{9 + 25\tan^2 x} \\ &= \int_0^1 \frac{du}{9 + 25u^2} \\ &= \frac{1}{25} \int_0^1 \frac{du}{\frac{9}{25} + u^2} \\ &= \frac{1}{25} \times \frac{5}{3} \left[\tan^{-1} \frac{5u}{3} \right]_0^1 \\ &= \frac{1}{15} \tan^{-1} \frac{5}{3} \end{aligned}$$

$$(i) (1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

$$\text{if } n=3, \quad 4^n = \binom{n}{0} + \binom{n}{1} \cdot 3 + \binom{n}{2}3^2 + \dots + \binom{n}{n}3^n$$

$$\text{now } \binom{n}{0}=1, \quad \therefore 3\binom{n}{1} + 3^2\binom{n}{2} + 3^3\binom{n}{3} + \dots + 3^n\binom{n}{n} = 4^n - 1$$

dividing by 3 and noting $2^{2n} = 4^n$

$$\binom{n}{1} + 3\binom{n}{2} + 9\binom{n}{3} + \dots + 3^{n-1}\binom{n}{n} = \frac{1}{3}(2^{2n} - 1)$$

Explain
divided around

not well done.

This line was
good. You
know $u = \tan x$
so you must
divide top and
bottom by $\cos^2 x$
(not hard
from here)

The increasing
powers of 3 mean

$$\begin{aligned} x &= 3. \\ 4^n &= 2^{2n} \\ \binom{n}{0} &= 1 \end{aligned}$$

$$(ii) \text{ if } x=1, \quad 2^n = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$$

$$\text{if } x=-1, \quad 0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n}$$

if n is even $(-1)^n$ is 1

$$\text{Adding:} \quad 2\binom{n}{0} + 2\binom{n}{2} + 2\binom{n}{4} + \dots + 2\binom{n}{n} = 2^n$$

$$\text{dividing by 2,} \quad \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots + \binom{n}{n} = 2^{n-1}$$

$$c) (i) \quad \binom{12}{4} = 495$$

$$(ii) \quad \binom{20}{4} - \binom{12}{4} = 4845 - 495 = 4350$$

(iii) Committee of 4 males must have a
male chairman: $\binom{8}{4} = 70$.

Committee of 3 males, female has $\frac{3}{4}$ chance
of a male chairman $\binom{8}{3} \times \binom{12}{1} \times \frac{3}{4} = 504$.

Committee of 2 males 2 females has $\frac{1}{2}$ chance
of a male chairman:

$$\binom{8}{2} \binom{12}{2} \times \frac{1}{2} = 924$$

Committee of 1 male 3 females has $\frac{1}{4}$ chance
of a male chairman

$$\binom{8}{1} \times \binom{12}{3} \times \frac{1}{4} = 440$$

$$\begin{aligned} \text{Total number} &= 70 + 504 + 924 + 440 \\ &= 1938. \end{aligned}$$

every second
term means
substitution
of $x=1$ and $x=-1$
and adding

Committees
are usually
combinations.